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Paper Code : PEC-IT602B Data Warehousing and Data Mining

UPID : 006584

Time Allotted : 3 Hours

Full Marks : 70

The Figures in the margin indicate full marks.

Candidate are required to give their answers in their own words as far as practicable

Group-A (Very Short Answer Type Question)

1. Answer any ten of the following :

[1 x 10 = 10]

- (i) The actual discovery phase of a knowledge discovery process is ____.
- (ii) What does OLTP stand for?
- (iii) Which is a class of learning algorithm that tries to find an optimum classification of a set of examples using the probabilistic theory?
- (iv) What is the reason of using Bayesian classifiers?
- (v) Is discrimination between spam and ham emails a classification task?
- (vi) What is the full form of KDD?
- (vii) Is volatility a property of the data warehouse?
- (viii) Measure of the probability that a certain hypothesis is incorrect given certain observations is ____ method.
- (ix) By maximizing the distances between nearest data point and hyper plane will help us to decide the right hyper-plane ____.
- (x) State the properties of Binary attribute?
- (xi) State whether True /False: Data warehouse is generally updated in real-time.
- (xii) Which is the process of subdivision of a set of examples into a number of classes?

Group-B (Short Answer Type Question)

Answer any three of the following :

[5 x 3 = 15]

2. When to use Logistic Regression and when to use Support Vector Machine? [5]
3. What is Cluster Analysis? [5]
4. What can we do with graph mining? [5]
5. What is Minimum support and Minimum confidence? [5]
6. What are time plot statistics? [5]

Group-C (Long Answer Type Question)

Answer any three of the following :

[15 x 3 = 45]

7. (a) Suppose the fraction of undergraduate students who smoke is 15% and the fraction of graduate students who smoke is 23%. If one-fifth of the college students are graduate students and the rest are undergraduates, what is the probability that a student who smokes is a graduate student? [8]
- (b) Given the information in part (a), is a randomly chosen college student more likely to be a graduate or undergraduate student? [7]
8. (a) What is Page Rank? [5]
- (b) What are Hubs and Authorities? [5]
- (c) What is HITS (Hyperlink Induced Topic Search)? [5]
9. (a) What is Web Structure Mining? [7]
- (b) Where can it be applied? [8]
10. Suppose we have market basket data consisting of 100 transactions and 20 items. If the support for item a is 25%, the support for item b is 90% and the support for itemset {a, b} is 20%. Let the support and confidence thresholds be 10% and 60%, respectively. [5+7+3]
- (a) Compute the confidence of the association rule {a} → {b}. Is the rule interesting according to the confidence measure?
- (b) Compute the interest measure for the association pattern {a, b}. Describe the nature of the relationship between item a and item b in terms of the interest measure.

(c) What conclusions can you draw from the results of parts (a) and (b)?

11. Consider the data set shown in Table

[8+7]

Record	A	B	C	Class
1	0	0	0	+
2	0	0	1	-
3	0	1	1	-
4	0	1	1	-
5	0	0	1	+
6	1	0	1	+
7	1	0	1	-
8	1	0	1	-
9	1	1	1	+
10	1	0	1	+

(a) Estimate the conditional probabilities for $P(A|+)$, $P(B|+)$, $P(C|+)$, $P(A|-)$, $P(B|-)$, and $P(C|-)$.

(b) Use the estimate of conditional probabilities given in the previous question to predict the class label for a test sample ($A = 0, B = 1, C = 0$) using the Naive Bayes approach.

*** END OF PAPER ***